Draft P&ID — review set

Décarbonation énergétique LPF & BSST – draft P&ID

Projet Décarbonation LPF & BSST · drawing PID-LPF-BSST-001 · revision A (draft for review) · generated 2026-10-23

DRAFT — NOT FOR CONSTRUCTION
Every addition is pending engineering review. Nothing was invented that the specification does not state.

What this is

A draft piping & instrumentation diagram generated from “Cahier des Charges P&ID Consolidé.pdf” (3 pages).

The AI read the specification into a traceable topology (every object points to a page and section); deterministic code then enriched, validated, laid out, drew and exported it.

Black items come from the specification. Orange items are proposals created by the project rules of section 5 of the specification and wait for approval.

Red numbered dots mark open points: values or choices the specification does not give. None of them was assumed.

Results of the automatic checks

Structural validation: PASS (0 errors, 0 warnings).

DEXPI Proteus XML round trip: PASS — 17 checks over 178 lines in 104 segments

Enrichment is idempotent: re-running it adds nothing (verified in this run).

The model in numbers

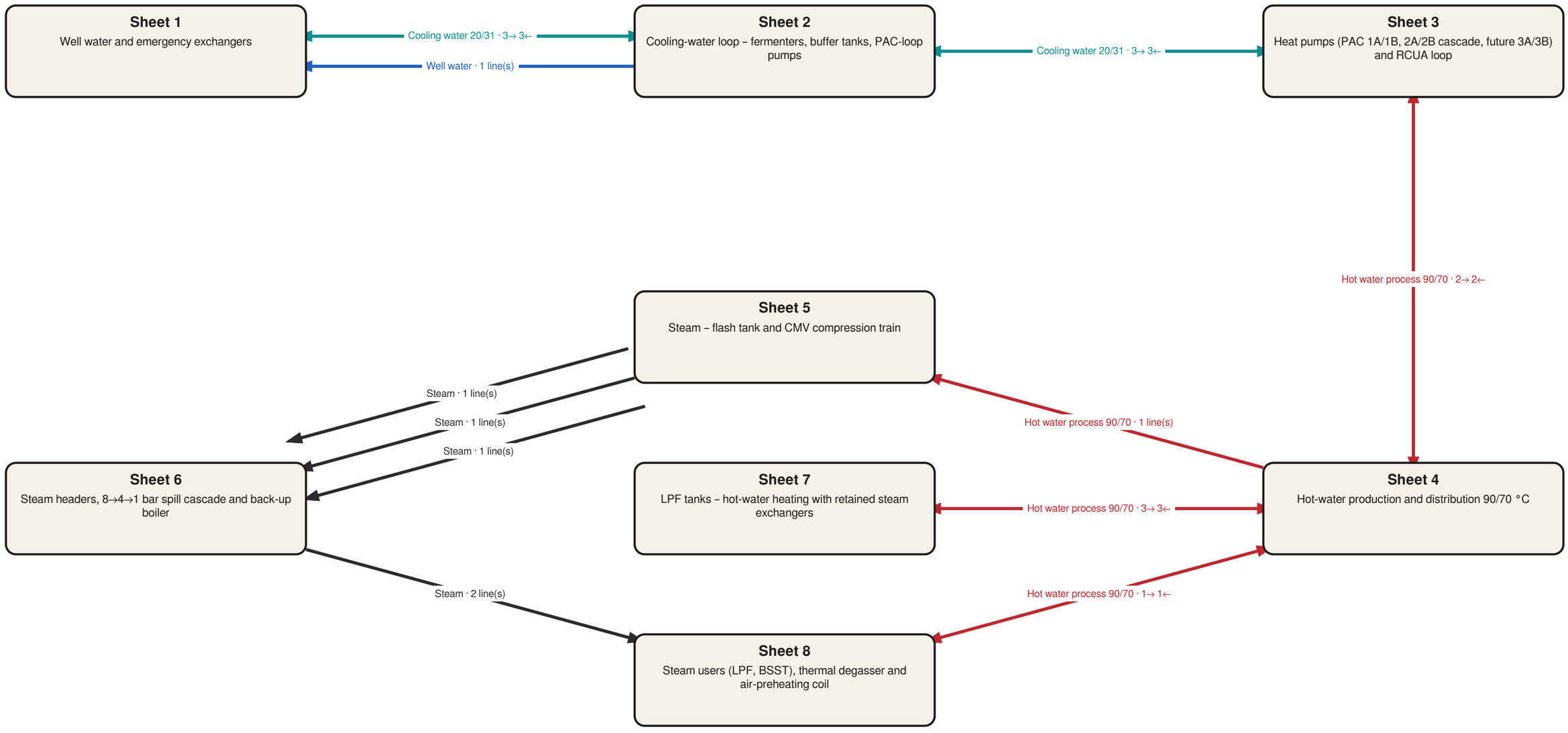
Equipment (source)	**45**
Process lines (after valve insertion)	**178**
Manifolds / tees	**38**
Valves (source / proposed)	**3 / 56**
Instruments (proposed)	**21**
Control loops	**1**
Proposals pending review	**181**
Open points	**24**
Interpretations to confirm	**68**

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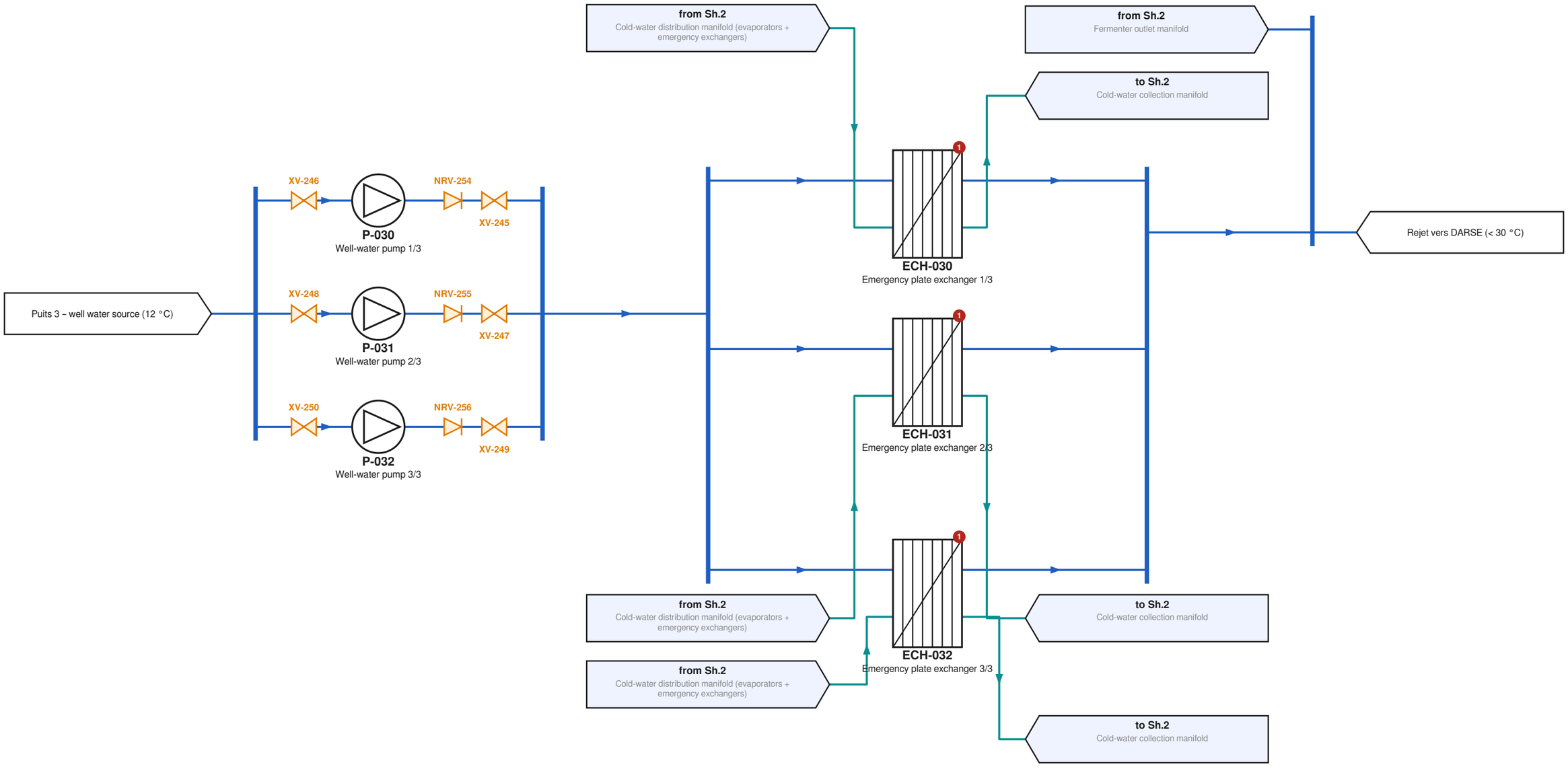
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Where an engineer should look first

1. Position of the emergency exchangers (ECH-030..032) on the cooling loop is ambiguous. §4 'Mode Normal' reads like a series path (through the exchangers, then to the PAC); §4 'Secours Échangeurs' (PAC isolated, exchangers cooling) and 'Secours Ultime' (valves closed toward PAC *and* exchangers) need them as a separate parallel branch. Modelled in PARALLEL with the evaporators.
2. Pumps P-033..035 are called 'Pompage Secondaire (Boucle PAC)'; no circulation driver is stated for the TK-001 → fermenters leg or for the hot-water loops (PAC condensers, 90 °C and RCUA distribution). Pumps were not invented; P-033..035 sit on the TK-002 outlet feeding the PAC/exchanger branches.
3. PAC 2A/2B: the table gives both units a 31/20 °C cold regime and 85/105 °C hot regime, but §2 routes the 2A condenser outlet (55 °C) into the 2B evaporator inlet. Modelled as a physical cascade: 2A evaporator on the cooling loop; 2A condenser ⇄ 2B evaporator via the 55/50 °C internal loop; only 2B condenser on the 85/105 °C RCUA loop. The 2B evaporator therefore has no cooling-loop connection, contrary to its table row.
9. Air-preheating coil: water inlet is limited to 85 °C (sanitary safety) while the process hot-water header is at 90 °C. No tempering/mixing arrangement is specified, so none is added. Only one example coil (TGE) is modelled; the number of coils per tower is not stated.
16. 'RCUA (85°C / 105°C)' is read as a single buffer tank. The PDF lists five buffer tanks (TK-001, TK-002, 70 °C, 90 °C, RCUA).



PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Overview – sheets and the services between them**	2 of 10	A	**DRAFT – pending engineering review**



LEGEND

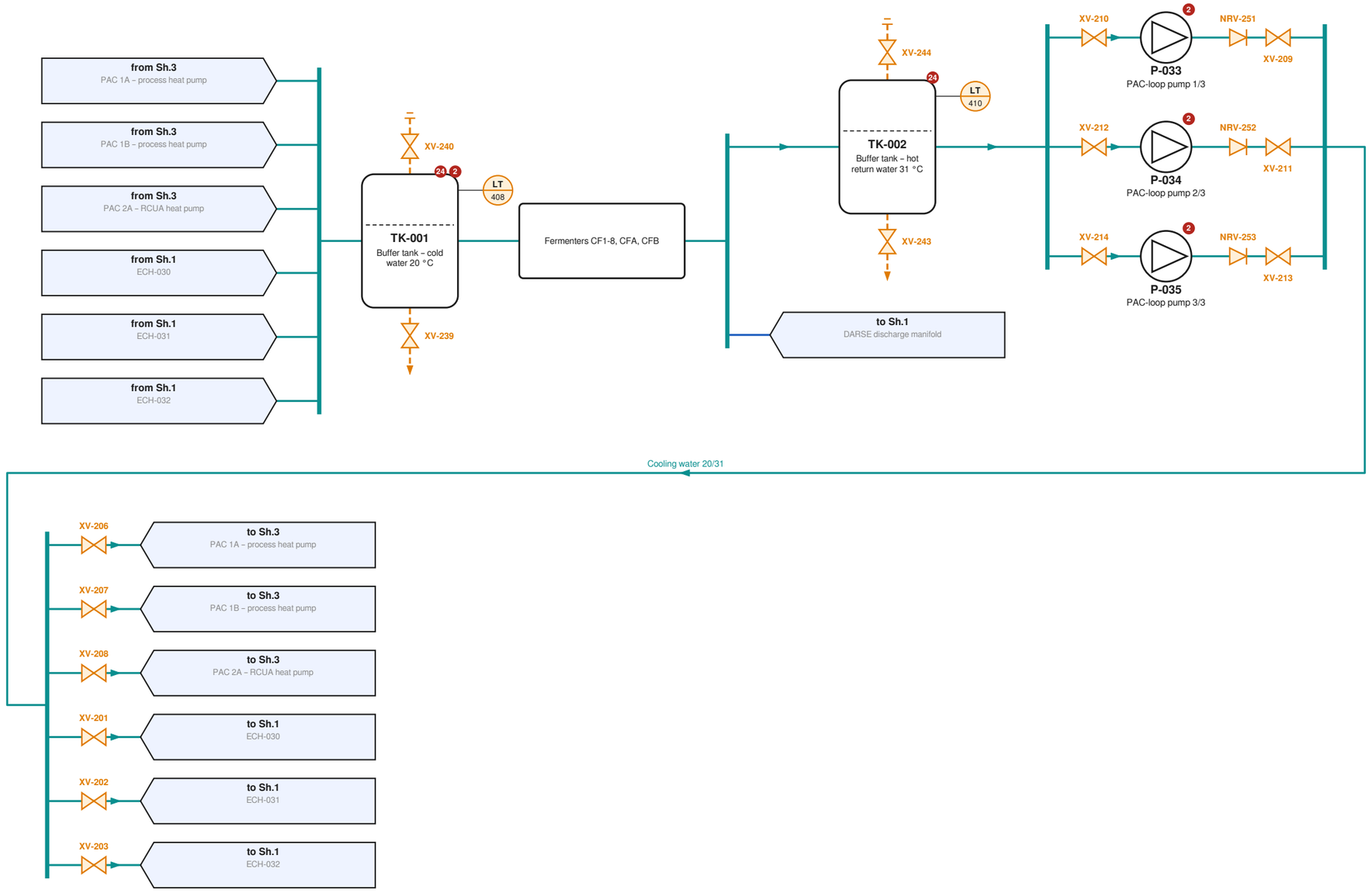
- Cooling water 20/31
- Well water
- Proposed addition – pending review
- (n) Open point – numbered list on the right

Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

OPEN POINTS ON THIS SHEET

1. Position of the emergency exchangers (ECH-030..032) on the cooling loop is ambiguous. §4 'Mode Normal' reads like a series path (through the exchangers, then to the PAC); §4 'Secours Échangeurs' (PAC isolated, exchangers cooling)…
21. Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Well water and emergency exchangers**	Sheet 1 of 8	A	**DRAFT – pending engineering review**



LEGEND

- Cooling water 20/31
- Drain
- Vent
- Well water

Proposed addition – pending review

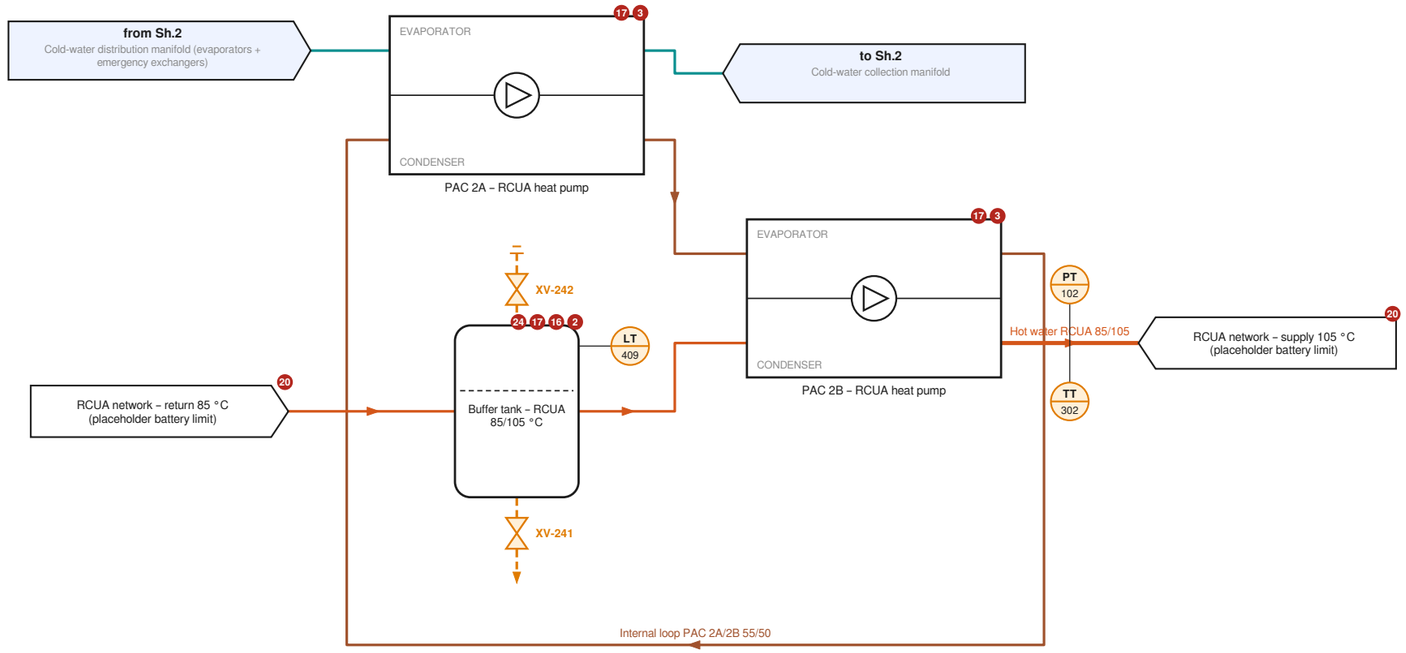
(n) Open point – numbered list on the right

Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

OPEN POINTS ON THIS SHEET

1. Position of the emergency exchangers (ECH-030..032) on the cooling loop is ambiguous. §4 'Mode Normal' reads like a series path (through the exchangers, then to the PAC); §4 'Secours Échangeurs' (PAC isolated, exchangers cooling)…
2. Pumps P-033..035 are called 'Pompage Secondaire (Boucle PAC)'; no circulation driver is stated for the TK-001 → fermenters leg or for the hot-water loops (PAC condensers, 90 °C and RCUA distribution). Pumps were not invented; P-033..035…
15. Ultimate mode: how well water reaches the fermenters (feed line) is not described; only the post-fermenter discharge to DARSE is.
21. Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.
24. Nozzles added for drain, vent and level tapping have no nozzle number and no size.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Cooling-water loop – fermenters, buffer tanks, PAC-loop pumps**	Sheet 2 of 8	A	**DRAFT – pending engineering review**



LEGEND

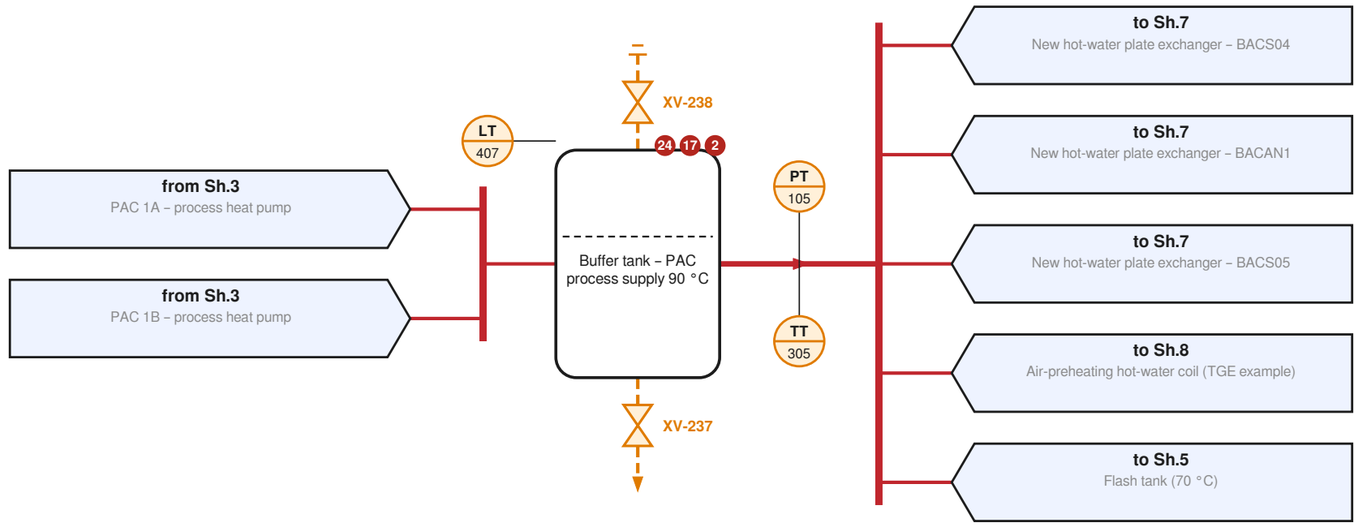
- Cooling water 20/31
- Drain
- Hot water process 90/70
- Hot water RCUA 85/105
- Internal loop PAC 2A/2B
- Vent
- Proposed addition – pending review
- Future equipment (LifeCyclePhase=Future)
- n Open point – numbered list on the right

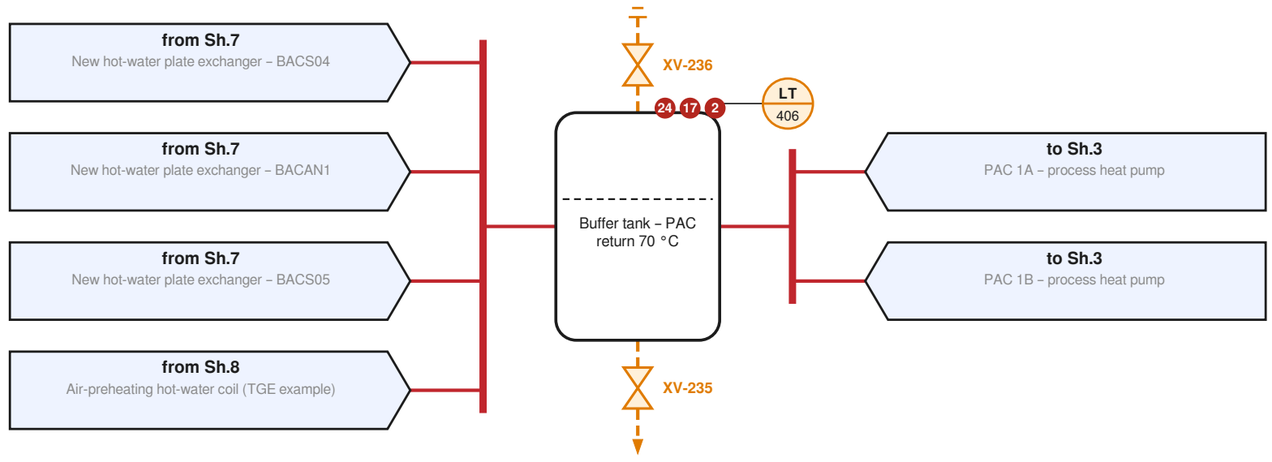
Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

OPEN POINTS ON THIS SHEET

- **2** Pumps P-033..035 are called 'Pompage Secondaire (Boucle PAC)'; no circulation driver is stated for the TK-001 → fermenters leg or for the hot-water loops (PAC condensers, 90 °C and RCUA distribution). Pumps were not invented; P-033..035…
- **3** PAC 2A/2B: the table gives both units a 31/20 °C cold regime and 85/105 °C hot regime, but §2 routes the 2A condenser outlet (55 °C) into the 2B evaporator inlet. Modelled as a physical cascade: 2A evaporator on the cooling loop; 2A condenser…
- **5** PAC 3A/3B are future equipment with no capacity, regime or connection data. They are drawn dashed (LifeCyclePhase=Future) and left unconnected.
- **16** 'RCUA (85°C / 105°C)' is read as a single buffer tank. The PDF lists five buffer tanks (TK-001, TK-002, 70 °C, 90 °C, RCUA).
- **17** Equipment tags are given only for P-030..035, ECH-030..032, TK-001/002, K3/MESSO/GEA and the LPF tanks. Heat pumps, 70/90/RCUA tanks, flash tank, CMV stages, boiler and the new exchangers have no tag.
- **20** RCUA network battery limits (destination of the 105 °C header and the 85 °C return) are not described; placeholder boundaries are used.
- **21** Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.
- **24** Nozzles added for drain, vent and level tapping have no nozzle number and no size.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Heat pumps (PAC 1A/1B, 2A/2B cascade, future 3A/3B) and RCUA loop**	Sheet 3 of 8	A	**DRAFT – pending engineering review**





LEGEND

- Drain
- Hot water process 90/70
- Vent

- Proposed addition – pending review
- (n) Open point – numbered list on the right

Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

OPEN POINTS ON THIS SHEET

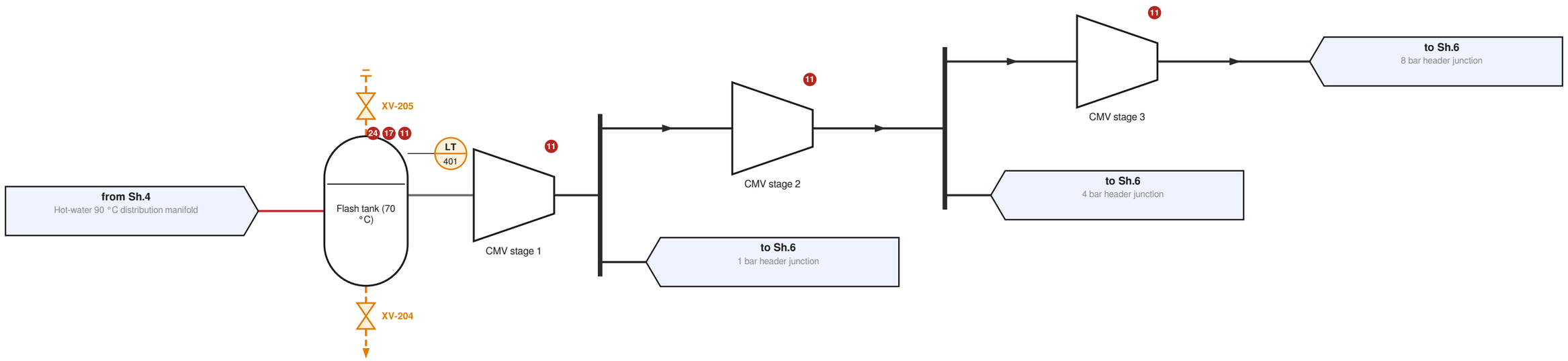
(2) Pumps P-033..035 are called 'Pompage Secondaire (Boucle PAC)'; no circulation driver is stated for the TK-001 → fermenters leg or for the hot-water loops (PAC condensers, 90 °C and RCUA distribution). Pumps were not invented; P-033..035…

(17) Equipment tags are given only for P-030..035, ECH-030..032, TK-001/002, K3/MESSO/GEA and the LPF tanks. Heat pumps, 70/90/RCUA tanks, flash tank, CMV stages, boiler and the new exchangers have no tag.

(21) Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.

(24) Nozzles added for drain, vent and level tapping have no nozzle number and no size.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Hot-water production and distribution 90/70 °C**	Sheet 4 of 8	A	**DRAFT – pending engineering review**



LEGEND

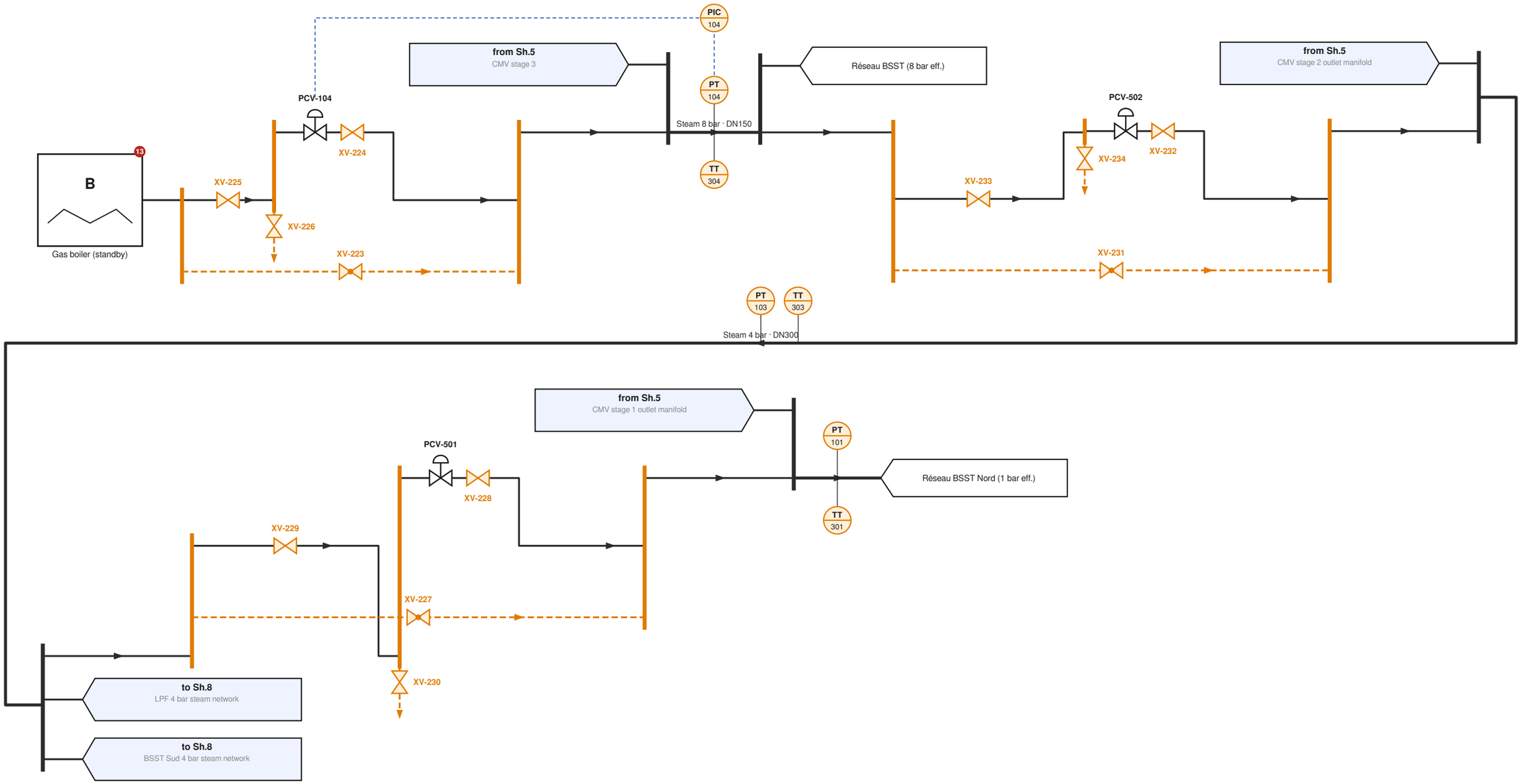
- Drain
- Flash steam
- Hot water process 90/70
- Steam
- Vent
- Proposed addition – pending review
- (n) Open point – numbered list on the right

Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

OPEN POINTS ON THIS SHEET

11. Flash tank: operating pressure and the destination of the 70 °C liquid are not stated; the liquid outlet is left open. The flash-vapour → CMV stage 1 and the inter-stage connections are INTERPRETED from '1 ligne ... sur 3 étages'.
17. Equipment tags are given only for P-030..035, ECH-030..032, TK-001/002, K3/MESSO/GEA and the LPF tanks. Heat pumps, 70/90/RCUA tanks, flash tank, CMV stages, boiler and the new exchangers have no tag.
21. Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.
24. Nozzles added for drain, vent and level tapping have no nozzle number and no size.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Steam – flash tank and CMV compression train**	Sheet 5 of 8	A	**DRAFT – pending engineering review**



LEGEND

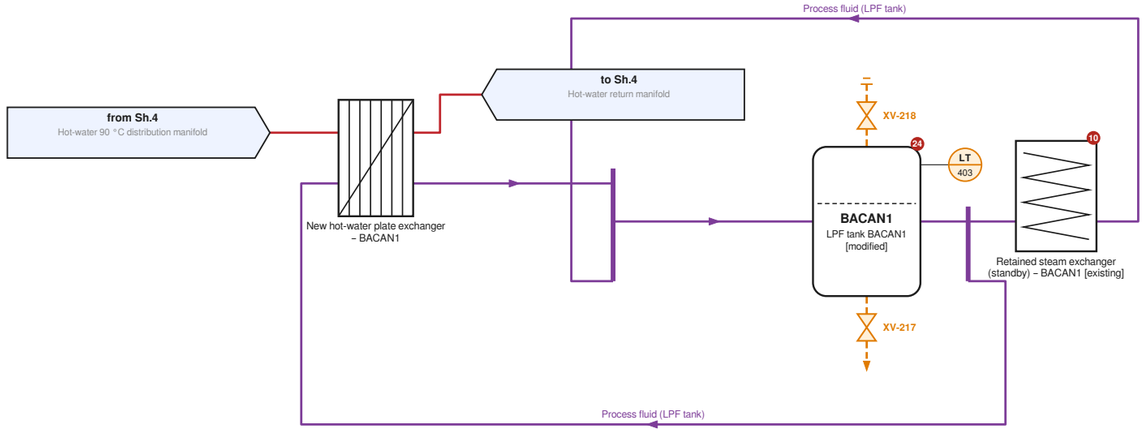
- Steam
- Proposed addition – pending review
- (n) Open point – numbered list on the right

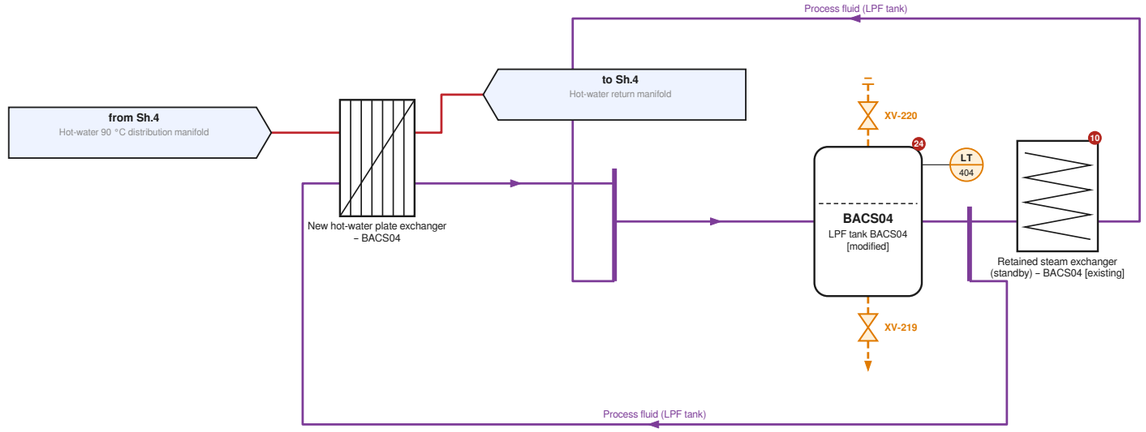
Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

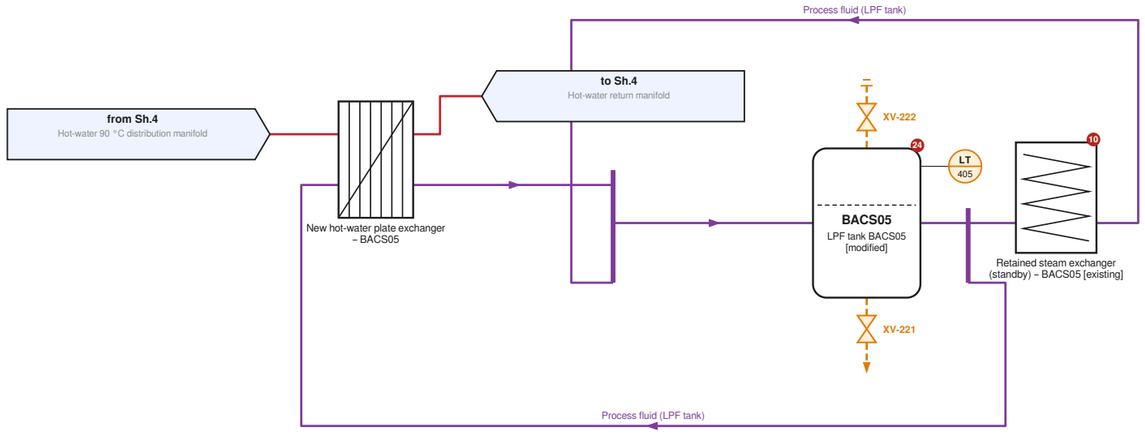
OPEN POINTS ON THIS SHEET

12. Spill valves 8 → 4 and 4 → 1 bar: controlled variable, measurement point, setpoint, valve technology and failure position are not stated, so no control loop is created for them.
13. Gas boiler: capacity, fuel/feed-water/flue connections, PCV technology and failure position are not stated.
21. Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.
22. Failure position (fail open / close / retain) of the control valves is not specified and was not assumed.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Steam headers, 8→4→1 bar spill cascade and back-up boiler**	Sheet 6 of 8	A	**DRAFT – pending engineering review**







LEGEND

- Drain
- Hot water process 90/70
- Process fluid (LPF tanks)
- Vent

- Proposed addition – pending review
- (n) Open point – numbered list on the right

Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

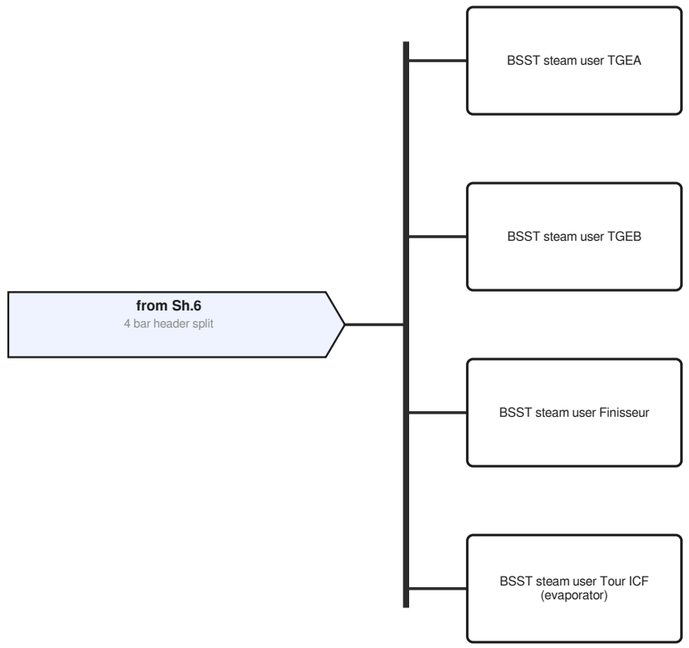
OPEN POINTS ON THIS SHEET

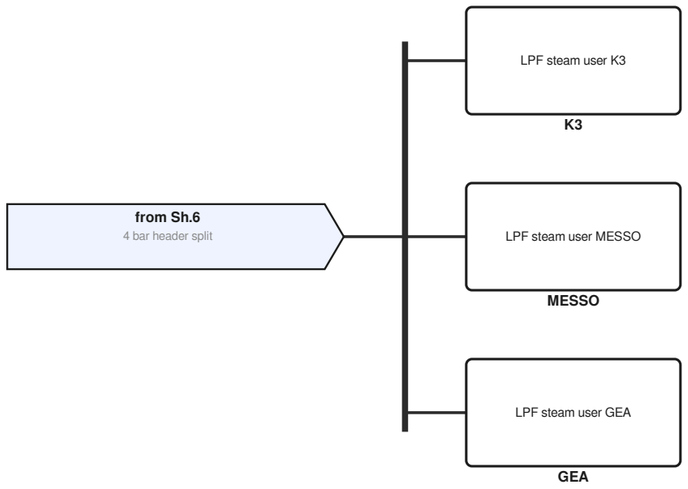
10 LPF tank heating loops: the circulation pump, fluid and steam supply of the retained steam exchangers are not described; their steam/condensate nozzles are left open.

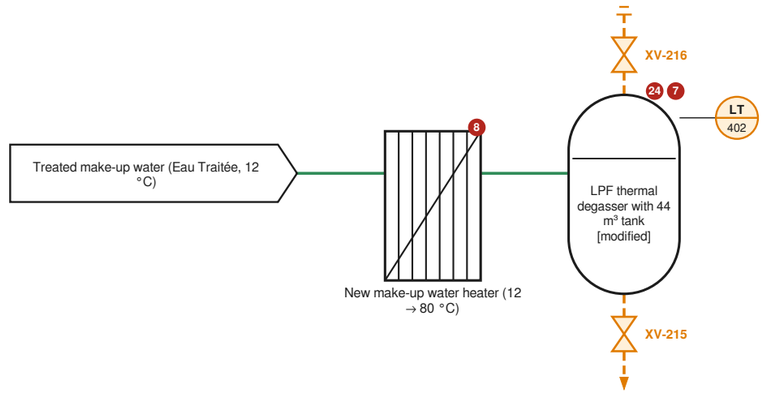
21 Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.

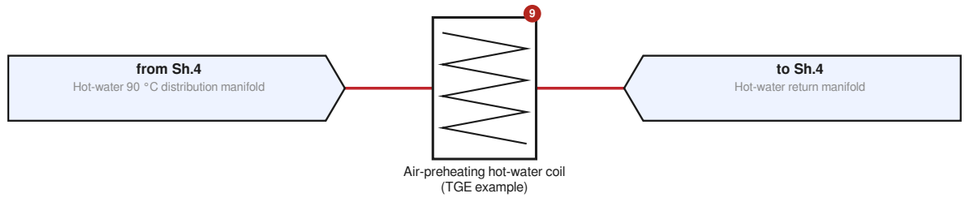
24 Nozzles added for drain, vent and level tapping have no nozzle number and no size.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**LPF tanks – hot-water heating with retained steam exchangers**	Sheet 7 of 8	A	**DRAFT – pending engineering review**









LEGEND

- Drain
- Hot water process 90/70
- Steam
- Treated make-up water
- Vent
- Proposed addition – pending review
- (n) Open point – numbered list on the right

Valves: gate/generic ▷◁ · check ▷| · globe ▷●◁ · control + actuator · bubble = function / proposed tag no.

OPEN POINTS ON THIS SHEET

7. Degasser steam supply: '4 bar ou 1 bar eff. (à définir)'. The steam inlet is left unconnected until chosen.
8. The heating medium and connection of the new make-up water heater (12 → 80 °C) are not stated; its hot side is left open.
9. Air-preheating coil: water inlet is limited to 85 °C (sanitary safety) while the process hot-water header is at 90 °C. No tempering/mixing arrangement is specified, so none is added. Only one example coil (TGE) is modelled; the number…
21. Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.
24. Nozzles added for drain, vent and level tapping have no nozzle number and no size.

PROJECT	DRAWING	SHEET	REV.	STATUS
Projet Décarbonation LPF & BSST	**Steam users (LPF, BSST), thermal degasser and air-preheating coil**	Sheet 8 of 8	A	**DRAFT – pending engineering review**

Proposed additions pending review

Everything below was created by a project rule of section 5 / section 4 of the specification. Approve or reject each line; tags are proposals (ISA 5.1, `tag_status: proposed`). Generic valves are listed with their technology as “to be selected”.

Tag (proposed)	Item	Rule	Why / where
PIC-104	Pressure indicating controller for cr.boiler_backup	control_loop.from_requirement	cr.boiler_backup states an automatic control action with an explicit setpoint (7.5 bar eff.); a controller is therefore required between measurement and final element.
—	Pressure loop for cr.boiler_backup	control_loop.from_requirement	loop = measurement → controller → final element for cr.boiler_backup.
XV-223	Bypass globe valve – Boiler back-up pressure control valve (PCV)	control_valve.manifold	bypass line of Boiler back-up pressure control valve (PCV) is equipped with a globe valve (technology explicit in the specification).
XV-224	Downstream isolation valve – Boiler back-up pressure control valve (PCV)	control_valve.manifold	control valve Boiler back-up pressure control valve (PCV) needs an downstream isolation valve. Technology not specified: generic valve.
XV-225	Upstream isolation valve – Boiler back-up pressure control valve (PCV)	control_valve.manifold	control valve Boiler back-up pressure control valve (PCV) needs an upstream isolation valve. Technology not specified: generic valve.
XV-226	Purge valve – Boiler back-up pressure control valve (PCV)	control_valve.manifold	control-valve manifold of Boiler back-up pressure control valve (PCV) includes a purge valve ('soupape de purge'). Technology not specified: generic valve.
XV-227	Bypass globe valve – Overproduction spill valve 4 → 1 bar (PCV)	control_valve.manifold	bypass line of Overproduction spill valve 4 → 1 bar (PCV) is equipped with a globe valve (technology explicit in the specification).
XV-228	Downstream isolation valve – Overproduction spill valve 4 → 1 bar (PCV)	control_valve.manifold	control valve Overproduction spill valve 4 → 1 bar (PCV) needs an downstream isolation valve. Technology not specified: generic valve.
XV-229	Upstream isolation valve – Overproduction spill valve 4 → 1 bar (PCV)	control_valve.manifold	control valve Overproduction spill valve 4 → 1 bar (PCV) needs an upstream isolation valve. Technology not specified: generic valve.
XV-230	Purge valve – Overproduction spill valve 4 → 1 bar (PCV)	control_valve.manifold	control-valve manifold of Overproduction spill valve 4 → 1 bar (PCV) includes a purge valve ('soupape de purge'). Technology not specified: generic valve.
XV-231	Bypass globe valve – Overproduction spill valve 8 → 4 bar (PCV)	control_valve.manifold	bypass line of Overproduction spill valve 8 → 4 bar (PCV) is equipped with a globe valve (technology explicit in the specification).
XV-232	Downstream isolation valve – Overproduction spill valve 8 → 4 bar (PCV)	control_valve.manifold	control valve Overproduction spill valve 8 → 4 bar (PCV) needs an downstream isolation valve. Technology not specified: generic valve.
XV-233	Upstream isolation valve – Overproduction spill valve 8 → 4 bar (PCV)	control_valve.manifold	control valve Overproduction spill valve 8 → 4 bar (PCV) needs an upstream isolation valve. Technology not specified: generic valve.
XV-234	Purge valve – Overproduction spill valve 8 → 4 bar (PCV)	control_valve.manifold	control-valve manifold of Overproduction spill valve 8 → 4 bar (PCV) includes a purge valve ('soupape de purge'). Technology not specified: generic valve.
—	Bypass merge – Boiler back-up pressure control valve (PCV) (tee)	control_valve.manifold	bypass return point downstream of Boiler back-up pressure control valve (PCV).
—	Bypass merge – Overproduction spill valve 4 → 1 bar (PCV) (tee)	control_valve.manifold	bypass return point downstream of Overproduction spill valve 4 → 1 bar (PCV).
—	Bypass merge – Overproduction spill valve 8 → 4 bar (PCV) (tee)	control_valve.manifold	bypass return point downstream of Overproduction spill valve 8 → 4 bar (PCV).
—	Bypass split – Boiler back-up pressure control valve (PCV) (tee)	control_valve.manifold	bypass branch point upstream of Boiler back-up pressure control valve (PCV).
—	Bypass split – Overproduction spill valve 4 → 1 bar (PCV) (tee)	control_valve.manifold	bypass branch point upstream of Overproduction spill valve 4 → 1 bar (PCV).
—	Bypass split – Overproduction spill valve 8 → 4 bar (PCV) (tee)	control_valve.manifold	bypass branch point upstream of Overproduction spill valve 8 → 4 bar (PCV).
—	Purge tee – Boiler back-up pressure control valve (PCV) (tee)	control_valve.manifold	purge branch on the control-valve body side of Boiler back-up pressure control valve (PCV).
—	Purge tee – Overproduction spill valve 4 → 1 bar (PCV) (tee)	control_valve.manifold	purge branch on the control-valve body side of Overproduction spill valve 4 → 1 bar (PCV).
—	Purge tee – Overproduction spill valve 8 → 4 bar (PCV) (tee)	control_valve.manifold	purge branch on the control-valve body side of Overproduction spill valve 8 → 4 bar (PCV).
PT-101	Pressure transmitter – Steam 1 bar header	header.instrumentation	main header 'Steam 1 bar' (ln.h1) must carry a PT.
PT-102	Pressure transmitter – Hot water RCUA 85/105 header	header.instrumentation	main header 'Hot water RCUA 85/105' (ln.h105) must carry a PT.
PT-103	Pressure transmitter – Steam 4 bar header	header.instrumentation	main header 'Steam 4 bar' (ln.h4) must carry a PT.
PT-104	Pressure transmitter – Steam 8 bar header	header.instrumentation	main header 'Steam 8 bar' (ln.h8) must carry a PT.
PT-105	Pressure transmitter – Hot water 90/70 header	header.instrumentation	main header 'Hot water 90/70' (ln.hw90) must carry a PT.
TT-301	Temperature transmitter – Steam 1 bar header	header.instrumentation	main header 'Steam 1 bar' (ln.h1) must carry a TT.
TT-302	Temperature transmitter – Hot water RCUA 85/105 header	header.instrumentation	main header 'Hot water RCUA 85/105' (ln.h105) must carry a TT.
TT-303	Temperature transmitter – Steam 4 bar header	header.instrumentation	main header 'Steam 4 bar' (ln.h4) must carry a TT.
TT-304	Temperature transmitter – Steam 8 bar header	header.instrumentation	main header 'Steam 8 bar' (ln.h8) must carry a TT.
TT-305	Temperature transmitter – Hot water 90/70 header	header.instrumentation	main header 'Hot water 90/70' (ln.hw90) must carry a TT.
XV-201	Isolation valve – ECH-030 hot_in	mode.isolation	mode 'mode.ultimate' requires isolation toward ECH-030 hot_in; no isolation valve exists there. Technology not specified: generic valve.
XV-202	Isolation valve – ECH-031 hot_in	mode.isolation	mode 'mode.ultimate' requires isolation toward ECH-031 hot_in; no isolation valve exists there. Technology not specified: generic valve.
XV-203	Isolation valve – ECH-032 hot_in	mode.isolation	mode 'mode.ultimate' requires isolation toward ECH-032 hot_in; no isolation valve exists there. Technology not specified: generic valve.
XV-206	Isolation valve – PAC 1A – process heat pump evap_in	mode.isolation	mode 'mode.emergency_hx' requires isolation toward PAC 1A – process heat pump evap_in; no isolation valve exists there. Technology not specified: generic valve.
XV-207	Isolation valve – PAC 1B – process heat pump evap_in	mode.isolation	mode 'mode.emergency_hx' requires isolation toward PAC 1B – process heat pump evap_in; no isolation valve exists there. Technology not specified: generic valve.
XV-208	Isolation valve – PAC 2A – RCUA heat pump evap_in	mode.isolation	mode 'mode.emergency_hx' requires isolation toward PAC 2A – RCUA heat pump evap_in; no isolation valve exists there. Technology not specified: generic valve.
NRV-251	Check valve – P-033 discharge	pump.auxiliaries	P-033 discharge needs a check valve ('clapet anti-retour') after the pump.
NRV-252	Check valve – P-034 discharge	pump.auxiliaries	P-034 discharge needs a check valve ('clapet anti-retour') after the pump.
NRV-253	Check valve – P-035 discharge	pump.auxiliaries	P-035 discharge needs a check valve ('clapet anti-retour') after the pump.
NRV-254	Check valve – P-030 discharge	pump.auxiliaries	P-030 discharge needs a check valve ('clapet anti-retour') after the pump.
NRV-255	Check valve – P-031 discharge	pump.auxiliaries	P-031 discharge needs a check valve ('clapet anti-retour') after the pump.
NRV-256	Check valve – P-032 discharge	pump.auxiliaries	P-032 discharge needs a check valve ('clapet anti-retour') after the pump.
XV-209	Discharge isolation valve – P-033	pump.auxiliaries	P-033 discharge needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-210	Suction isolation valve – P-033	pump.auxiliaries	P-033 suction needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.

Tag (proposed)	Item	Rule	Why / where
XV-211	Discharge isolation valve – P-034	pump.auxiliaries	P-034 discharge needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-212	Suction isolation valve – P-034	pump.auxiliaries	P-034 suction needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-213	Discharge isolation valve – P-035	pump.auxiliaries	P-035 discharge needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-214	Suction isolation valve – P-035	pump.auxiliaries	P-035 suction needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-245	Discharge isolation valve – P-030	pump.auxiliaries	P-030 discharge needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-246	Suction isolation valve – P-030	pump.auxiliaries	P-030 suction needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-247	Discharge isolation valve – P-031	pump.auxiliaries	P-031 discharge needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-248	Suction isolation valve – P-031	pump.auxiliaries	P-031 suction needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-249	Discharge isolation valve – P-032	pump.auxiliaries	P-032 discharge needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
XV-250	Suction isolation valve – P-032	pump.auxiliaries	P-032 suction needs a manual isolation valve. Technology is not specified by the source, so a generic valve is used.
LT-401	Level transmitter – Flash tank (70 °C)	vessel.auxiliaries	capacitive equipment Flash tank (70 °C) needs a level transmitter.
LT-402	Level transmitter – LPF thermal degasser with 44 m³ tank	vessel.auxiliaries	capacitive equipment LPF thermal degasser with 44 m³ tank needs a level transmitter. Existing equipment: verify the already installed drain/vent/level device before accepting.
LT-403	Level transmitter – BACAN1	vessel.auxiliaries	capacitive equipment BACAN1 needs a level transmitter. Existing equipment: verify the already installed drain/vent/level device before accepting.
LT-404	Level transmitter – BACS04	vessel.auxiliaries	capacitive equipment BACS04 needs a level transmitter. Existing equipment: verify the already installed drain/vent/level device before accepting.
LT-405	Level transmitter – BACS05	vessel.auxiliaries	capacitive equipment BACS05 needs a level transmitter. Existing equipment: verify the already installed drain/vent/level device before accepting.
LT-406	Level transmitter – Buffer tank – PAC return 70 °C	vessel.auxiliaries	capacitive equipment Buffer tank – PAC return 70 °C needs a level transmitter.
LT-407	Level transmitter – Buffer tank – PAC process supply 90 °C	vessel.auxiliaries	capacitive equipment Buffer tank – PAC process supply 90 °C needs a level transmitter.
LT-408	Level transmitter – TK-001	vessel.auxiliaries	capacitive equipment TK-001 needs a level transmitter.
LT-409	Level transmitter – Buffer tank – RCUA 85/105 °C	vessel.auxiliaries	capacitive equipment Buffer tank – RCUA 85/105 °C needs a level transmitter.
LT-410	Level transmitter – TK-002	vessel.auxiliaries	capacitive equipment TK-002 needs a level transmitter.
XV-204	Drain valve – Flash tank (70 °C)	vessel.auxiliaries	capacitive equipment Flash tank (70 °C) needs a low-point drain valve. Technology not specified: generic valve.
XV-205	Vent valve – Flash tank (70 °C)	vessel.auxiliaries	capacitive equipment Flash tank (70 °C) needs a high-point vent valve. Technology not specified: generic valve.
XV-215	Drain valve – LPF thermal degasser with 44 m³ tank	vessel.auxiliaries	capacitive equipment LPF thermal degasser with 44 m³ tank needs a low-point drain valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-216	Vent valve – LPF thermal degasser with 44 m³ tank	vessel.auxiliaries	capacitive equipment LPF thermal degasser with 44 m³ tank needs a high-point vent valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-217	Drain valve – BACAN1	vessel.auxiliaries	capacitive equipment BACAN1 needs a low-point drain valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-218	Vent valve – BACAN1	vessel.auxiliaries	capacitive equipment BACAN1 needs a high-point vent valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-219	Drain valve – BACS04	vessel.auxiliaries	capacitive equipment BACS04 needs a low-point drain valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-220	Vent valve – BACS04	vessel.auxiliaries	capacitive equipment BACS04 needs a high-point vent valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-221	Drain valve – BACS05	vessel.auxiliaries	capacitive equipment BACS05 needs a low-point drain valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-222	Vent valve – BACS05	vessel.auxiliaries	capacitive equipment BACS05 needs a high-point vent valve. Existing equipment: verify the already installed drain/vent/level device before accepting. Technology not specified: generic valve.
XV-235	Drain valve – Buffer tank – PAC return 70 °C	vessel.auxiliaries	capacitive equipment Buffer tank – PAC return 70 °C needs a low-point drain valve. Technology not specified: generic valve.
XV-236	Vent valve – Buffer tank – PAC return 70 °C	vessel.auxiliaries	capacitive equipment Buffer tank – PAC return 70 °C needs a high-point vent valve. Technology not specified: generic valve.
XV-237	Drain valve – Buffer tank – PAC process supply 90 °C	vessel.auxiliaries	capacitive equipment Buffer tank – PAC process supply 90 °C needs a low-point drain valve. Technology not specified: generic valve.
XV-238	Vent valve – Buffer tank – PAC process supply 90 °C	vessel.auxiliaries	capacitive equipment Buffer tank – PAC process supply 90 °C needs a high-point vent valve. Technology not specified: generic valve.
XV-239	Drain valve – TK-001	vessel.auxiliaries	capacitive equipment TK-001 needs a low-point drain valve. Technology not specified: generic valve.
XV-240	Vent valve – TK-001	vessel.auxiliaries	capacitive equipment TK-001 needs a high-point vent valve. Technology not specified: generic valve.
XV-241	Drain valve – Buffer tank – RCUA 85/105 °C	vessel.auxiliaries	capacitive equipment Buffer tank – RCUA 85/105 °C needs a low-point drain valve. Technology not specified: generic valve.
XV-242	Vent valve – Buffer tank – RCUA 85/105 °C	vessel.auxiliaries	capacitive equipment Buffer tank – RCUA 85/105 °C needs a high-point vent valve. Technology not specified: generic valve.
XV-243	Drain valve – TK-002	vessel.auxiliaries	capacitive equipment TK-002 needs a low-point drain valve. Technology not specified: generic valve.
XV-244	Vent valve – TK-002	vessel.auxiliaries	capacitive equipment TK-002 needs a high-point vent valve. Technology not specified: generic valve.

Changes to source objects (source value preserved)

Object	Field	Source value	Proposed	Rule
eq.tank_cold	nozzles	['inlet', 'outlet']	**['inlet', 'outlet', 'drain']**	vessel.auxiliaries
eq.tank_cold	nozzles	['inlet', 'outlet', 'drain']	**['inlet', 'outlet', 'drain', 'vent']**	vessel.auxiliaries
eq.tank_cold	nozzles	['inlet', 'outlet', 'drain', 'vent']	**['inlet', 'outlet', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.tank_warm	nozzles	['inlet', 'outlet']	**['inlet', 'outlet', 'drain']**	vessel.auxiliaries
eq.tank_warm	nozzles	['inlet', 'outlet', 'drain']	**['inlet', 'outlet', 'drain', 'vent']**	vessel.auxiliaries
eq.tank_warm	nozzles	['inlet', 'outlet', 'drain', 'vent']	**['inlet', 'outlet', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries

Object	Field	Source value	Proposed	Rule
eq.tank_70	nozzles	['inlet', 'outlet']	**['inlet', 'outlet', 'drain']**	vessel.auxiliaries
eq.tank_70	nozzles	['inlet', 'outlet', 'drain']	**['inlet', 'outlet', 'drain', 'vent']**	vessel.auxiliaries
eq.tank_70	nozzles	['inlet', 'outlet', 'drain', 'vent']	**['inlet', 'outlet', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.tank_90	nozzles	['inlet', 'outlet']	**['inlet', 'outlet', 'drain']**	vessel.auxiliaries
eq.tank_90	nozzles	['inlet', 'outlet', 'drain']	**['inlet', 'outlet', 'drain', 'vent']**	vessel.auxiliaries
eq.tank_90	nozzles	['inlet', 'outlet', 'drain', 'vent']	**['inlet', 'outlet', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.tank_rcua	nozzles	['inlet', 'outlet']	**['inlet', 'outlet', 'drain']**	vessel.auxiliaries
eq.tank_rcua	nozzles	['inlet', 'outlet', 'drain']	**['inlet', 'outlet', 'drain', 'vent']**	vessel.auxiliaries
eq.tank_rcua	nozzles	['inlet', 'outlet', 'drain', 'vent']	**['inlet', 'outlet', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.flash_tank	nozzles	['hot_water_in', 'steam_out', 'liquid_out']	**['hot_water_in', 'steam_out', 'liquid_out', 'drain']**	vessel.auxiliaries
eq.flash_tank	nozzles	['hot_water_in', 'steam_out', 'liquid_out', 'drain']	**['hot_water_in', 'steam_out', 'liquid_out', 'drain', 'vent']**	vessel.auxiliaries
eq.flash_tank	nozzles	['hot_water_in', 'steam_out', 'liquid_out', 'drain', 'vent']	**['hot_water_in', 'steam_out', 'liquid_out', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.lpf_tank_s04	nozzles	['proc_out', 'proc_in']	**['proc_out', 'proc_in', 'drain']**	vessel.auxiliaries
eq.lpf_tank_s04	nozzles	['proc_out', 'proc_in', 'drain']	**['proc_out', 'proc_in', 'drain', 'vent']**	vessel.auxiliaries
eq.lpf_tank_s04	nozzles	['proc_out', 'proc_in', 'drain', 'vent']	**['proc_out', 'proc_in', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.lpf_tank_an1	nozzles	['proc_out', 'proc_in']	**['proc_out', 'proc_in', 'drain']**	vessel.auxiliaries
eq.lpf_tank_an1	nozzles	['proc_out', 'proc_in', 'drain']	**['proc_out', 'proc_in', 'drain', 'vent']**	vessel.auxiliaries
eq.lpf_tank_an1	nozzles	['proc_out', 'proc_in', 'drain', 'vent']	**['proc_out', 'proc_in', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.lpf_tank_s05	nozzles	['proc_out', 'proc_in']	**['proc_out', 'proc_in', 'drain']**	vessel.auxiliaries
eq.lpf_tank_s05	nozzles	['proc_out', 'proc_in', 'drain']	**['proc_out', 'proc_in', 'drain', 'vent']**	vessel.auxiliaries
eq.lpf_tank_s05	nozzles	['proc_out', 'proc_in', 'drain', 'vent']	**['proc_out', 'proc_in', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
eq.lpf_degasser	nozzles	['water_in', 'steam_in']	**['water_in', 'steam_in', 'drain']**	vessel.auxiliaries
eq.lpf_degasser	nozzles	['water_in', 'steam_in', 'drain']	**['water_in', 'steam_in', 'drain', 'vent']**	vessel.auxiliaries
eq.lpf_degasser	nozzles	['water_in', 'steam_in', 'drain', 'vent']	**['water_in', 'steam_in', 'drain', 'vent', 'level_tap']**	vessel.auxiliaries
pc.pcv_boiler	tag	None	**PCV-104**	tagging.isa
pc.pcv_spill_4_1	tag	None	**PCV-501**	tagging.isa
pc.pcv_spill_8_4	tag	None	**PCV-502**	tagging.isa

39 source lines were split to insert components; each child line keeps the service and properties of its source line (field `split_from`).

Open points – what the specification does not say

These are preserved as unresolved requirements. Nothing here was assumed (design pressures, relief sizing, safety thresholds, failure positions, material classes, nozzle sizes and instrument ranges are never invented).

#	Id	Kind	Open point	Affects	Question for the engineer
1	ur.ech_position	ambiguity	Position of the emergency exchangers (ECH-030..032) on the cooling loop is ambiguous. §4 'Mode Normal' reads like a series path (through the exchangers, then to the PAC); §4 'Secours Échangeurs' (PAC isolated, exchangers cooling) and 'Secours Ultime' (valves closed toward PAC *and* exchangers) need them as a separate parallel branch. Modelled in PARALLEL with the evaporators.	eq.emerg_hx_a, eq.emerg_hx_b, eq.emerg_hx_c, mf.cw_s, mf.cw_m	Confirm parallel arrangement and the well-water path in Normal mode.
2	ur.loop_pump_position	ambiguity	Pumps P-033..035 are called 'Pompage Secondaire (Boucle PAC)'; no circulation driver is stated for the TK-001 → fermenters leg or for the hot-water loops (PAC condensers, 90 °C and RCUA distribution). Pumps were not invented; P-033..035 sit on the TK-002 outlet feeding the PAC/exchanger branches.	eq.loop_pump_a, eq.loop_pump_b, eq.loop_pump_c, eq.tank_cold, eq.tank_70, eq.tank_90 … (+1)	Where are the circulation pumps of the fermenter leg and of the hot-water loops?
3	ur.cascade_2a_2b	conflict	PAC 2A/2B: the table gives both units a 31/20 °C cold regime and 85/105 °C hot regime, but §2 routes the 2A condenser outlet (55 °C) into the 2B evaporator inlet. Modelled as a physical cascade: 2A evaporator on the cooling loop; 2A condenser ⇄ 2B evaporator via the 55/50 °C internal loop; only 2B condenser on the 85/105 °C RCUA loop. The 2B evaporator therefore has no cooling-loop connection, contrary to its table row.	eq.hp_2a, eq.hp_2b, ln.pac2_internal	Confirm the 2A/2B cascade connections.
4	ur.internal_loop	missing_value	Internal loop PAC 2A/2B: nominal diameter is 'À définir'; the 50 °C return leg is implied, not stated.	ln.pac2_internal	DN of the internal loop and confirmation of its return leg.
5	ur.future_hp	scope	PAC 3A/3B are future equipment with no capacity, regime or connection data. They are drawn dashed (LifeCyclePhase=Future) and left unconnected.	eq.hp_3a, eq.hp_3b	
6	ur.hw_dn	missing_value	Hot-water process lines: only a range 'DN 200 à DN 350' is given; per-line diameters are not defined. Branch lines of the cooling-water loop carry only the 'DN 600 minimum' of the main lines.	ln.hw90	Line-by-line DN for hot-water and cooling-water branches.
7	ur.degasser_steam	selection	Degasser steam supply: '4 bar ou 1 bar eff. (à définir)'. The steam inlet is left unconnected until chosen.	eq.lpf_degasser	Select 4 bar or 1 bar for the degasser.
8	ur.degasser_hx_medium	missing_value	The heating medium and connection of the new make-up water heater (12 → 80 °C) are not stated; its hot side is left open.	eq.lpf_degasser_hx	Heating medium of the make-up heater.
9	ur.tge_coil	conflict	Air-preheating coil: water inlet is limited to 85 °C (sanitary safety) while the process hot-water header is at 90 °C. No tempering/mixing arrangement is specified, so none is added. Only one example coil (TGE) is modelled; the number of coils per tower is not stated.	eq.bsst_tge_coil, ln.hw90	How is the 85 °C inlet limit achieved, and how many coils are needed?
10	ur.lpf_tank_loops	missing_value	LPF tank heating loops: the circulation pump, fluid and steam supply of the retained steam exchangers are not described; their steam/condensate nozzles are left open.	eq.lpf_shx_s04, eq.lpf_shx_an1, eq.lpf_shx_s05	
11	ur.flash_tank	missing_value	Flash tank: operating pressure and the destination of the 70 °C liquid are not stated; the liquid outlet is left open. The flash-vapour → CMV stage 1 and the inter-stage connections are INTERPRETED from '1 ligne ... sur 3 étages'.	eq.flash_tank, eq.cmv_s1, eq.cmv_s2, eq.cmv_s3	Flash-tank liquid destination and pressure; confirm series staging.
12	ur.spill_control	missing_value	Spill valves 8 → 4 and 4 → 1 bar: controlled variable, measurement point, setpoint, valve technology and failure position are not stated, so no control loop is created for them.	pc.pcv_spill_8_4, pc.pcv_spill_4_1, cr.spill_8_4, cr.spill_4_1	Control philosophy of the overproduction cascade.
13	ur.boiler	missing_value	Gas boiler: capacity, fuel/feed-water/flue connections, PCV technology and failure position are not stated.	eq.boiler, pc.pcv_boiler	
14	ur.hw_backup	scope	'Les réseaux Eau Chaude basculent sur le secours vapeur local' (emergency-exchanger mode): the local steam back-up equipment is not described.	mode.emergency_hx	
15	ur.ultimate_feed	missing_value	Ultimate mode: how well water reaches the fermenters (feed line) is not described; only the post-fermenter discharge to DARSE is.	mode.ultimate, mf.ferm_out	
16	ur.tank_count	ambiguity	'RCUA (85°C / 105°C)' is read as a single buffer tank. The PDF lists five buffer tanks (TK-001, TK-002, 70 °C, 90 °C, RCUA).	eq.tank_rcua	One RCUA tank or one per temperature level?
17	ur.tags	missing_value	Equipment tags are given only for P-030..035, ECH-030..032, TK-001/002, K3/MESSO/GEA and the LPF tanks. Heat pumps, 70/90/RCUA tanks, flash tank, CMV stages, boiler and the new exchangers have no tag.	eq.hp_1a, eq.hp_1b, eq.hp_2a, eq.hp_2b, eq.tank_70, eq.tank_90 … (+2)	
18	ur.nozzles	missing_value	No nozzle numbers or nozzle sizes are given for any equipment (except DN50 on the TGE coil). Nozzle names are schematic.		Nozzle schedule.
19	ur.design_data	missing_value	No design pressures, relief devices/sizing, safety thresholds (other than 7.5 bar eff.), valve failure positions, material classes or instrument ranges are given. None are invented.		
20	ur.rcua_limits	scope	RCUA network battery limits (destination of the 105 °C header and the 85 °C return) are not described; placeholder boundaries are used.	bc.rcua_out, bc.rcua_return	
21	ur.gen.valve_technology	selection	Valve technology is not specified for these valves. They use the explicitly identified generic DEXPI representation (CustomOperatedValve, 'technology to be selected'); no specific valve type was chosen.	pc.gen.emerg_hx_a.hot_in_isolation, pc.gen.emerg_hx_b.hot_in_isolation, pc.gen.emerg_hx_c.hot_in_isolation, pc.gen.flash_tank.drain, pc.gen.flash_tank.vent, pc.gen.hp_1a.evap_in_isolation … (+44)	Select the valve technology for each valve function.
22	ur.gen.fail_position	missing_value	Failure position (fail open / close / retain) of the control valves is not specified and was not assumed.	pc.pcv_boiler, pc.pcv_spill_4_1, pc.pcv_spill_8_4	
23	ur.gen.instrument_ranges	missing_value	Measuring ranges and alarm/trip settings of the generated instruments are not specified and were not assumed.	in.boiler_backup.pic, in.flash_tank.lt, in.h1.pt, in.h1.tt, in.h105.pt, in.h105.tt … (+15)	
24	ur.gen.nozzles	missing_value	Nozzles added for drain, vent and level tapping have no nozzle number and no size.	eq.flash_tank, eq.lpf_degasser, eq.lpf_tank_an1, eq.lpf_tank_s04, eq.lpf_tank_s05, eq.tank_70 … (+4)	

Interpretations to confirm

Facts that are not stated word for word in the specification but were needed to draw a connected diagram. They are in the source topology with `basis: interpreted` and the note below.

Objects	Page / section	Interpretation
4	p.1 · 1.1 Zone de Captage et Boucle Fermenteurs	Well water is the cooling medium of the emergency exchangers (§4: 'les échangeurs de secours refroidissent l'eau des fermenteurs').
1	p.1 · 1.2 Zone de Production de Chaleur (PAC) et Stockage	'RCUA (85°C / 105°C)' read as ONE tank covering both temperatures of the RCUA loop (see ur.tank_count).
1	p.1 · 1.2 Zone de Production de Chaleur (PAC) et Stockage	Placeholder battery limit; the RCUA network itself is not described.
3	p.1 · 1.2 Zone de Production de Chaleur (PAC) et Stockage	“PAC RCUA (2A / 2B) ... 85°C / 105°C”
4	p.1 · 1.3 Zone de Production de Vapeur (CMV)	Three stages of one line are read as a series compression train (stage n outlet feeds stage n+1 inlet).
1	p.1 · 1.3 Zone de Production de Vapeur (CMV)	Vapour from the flash tank is the suction of CMV stage 1.
1	p.1 · 1.3 Zone de Production de Vapeur (CMV)	“Alimentée par l'eau chaude à 90°C des PAC via un Flash Tank (70°C)”
3	p.2 · 2. Lignes principales et réseaux de distribution	Condenser outlets (90 °C) collected in the 90 °C departure tank.
1	p.2 · 2. Lignes principales et réseaux de distribution	Emergency exchangers placed in PARALLEL with the evaporators (see ur.ech_position).
3	p.2 · 2. Lignes principales et réseaux de distribution	PAC process condensers fed from the 70 °C return tank.
1	p.2 · 2. Lignes principales et réseaux de distribution	Pumps P-033..035 are the 'Pompage Secondaire (Boucle PAC)'; their position on the loop is not stated (see ur.loop_pump_position).
1	p.2 · 2. Lignes principales et réseaux de distribution	Return leg of the internal loop is implied by the 55/50 °C regime, not drawn in the text.
5	p.2 · 2. Lignes principales et réseaux de distribution	Return side of the 90/70 °C regime, collected in the 70 °C tank.
7	p.2 · 2. Lignes principales et réseaux de distribution	“Eau de refroidissement DN 600 (minimum) Aller 20°C / Retour 31°C PAC (Évaporateurs) <-> Fermenteurs”
4	p.2 · 3. Consommateurs process et modifications	Users listed under BSST are supplied from the BSST Sud 4 bar network.
3	p.2 · 3. Consommateurs process et modifications	Users listed under LPF are supplied from the LPF 4 bar network.
1	p.2 · 3. Maintien en température (Cuves LPF)	Existing steam exchanger kept in parallel as backup for BACAN1.
1	p.2 · 3. Maintien en température (Cuves LPF)	Existing steam exchanger kept in parallel as backup for BACS04.
1	p.2 · 3. Maintien en température (Cuves LPF)	Existing steam exchanger kept in parallel as backup for BACS05.
1	p.2 · 3. Maintien en température (Cuves LPF)	One new plate exchanger per listed LPF tank (BACAN1).
1	p.2 · 3. Maintien en température (Cuves LPF)	One new plate exchanger per listed LPF tank (BACS04).
1	p.2 · 3. Maintien en température (Cuves LPF)	One new plate exchanger per listed LPF tank (BACS05).
18	p.2 · 3. Maintien en température (Cuves LPF)	Process side of the new and the retained exchanger in parallel; the circulation arrangement is not described.
1	p.3 · 5. Instrumentation des Collecteurs (Headers)	The PDF names a 105 °C hot-water header but not its destination; placeholder boundary.